Time Series: Forecasting, Events, and Stationarity

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The Two Questions of This Course

> Given a time series, we want to answer two questions: what will happen next? and when did something relevant happen or is it happening?

Prediction

Estimate future values using temporal dependence. The past structures the future — models learn patterns to anticipate what has not yet been observed.

Event Detection

Identify moments or intervals where behavior deviates, changes, or repeats in a relevant way. Events are signals that break or confirm patterns.

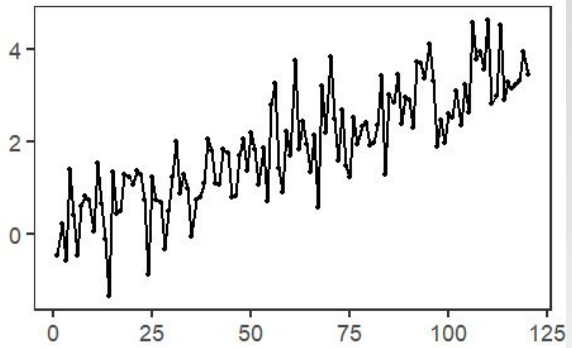
All the fundamentals of this course — mean, variance, autocorrelation, stationarity — serve these two tasks.

Time Series as a Sequence of Observations

A time series is an ordered sequence of values over time, where each observation is associated with a specific time instant. Order matters: the past informs the future and defines the context in which events occur.

$$x_1, x_2, \ldots, x_T$$

Where x_t is the value observed at time t, and T is the total number of observations. The series can also be represented as a function of time: $x(t)$.



Original Sequence

The natural order preserves the temporal structure: each value carries the history of what came before and conditions what may come next.

After Permutation

Shuffling the order destroys the temporal structure. Unlike ordinary data, we cannot rearrange the values without altering the meaning of the information.

In traditional data, order does not matter; in time series, it is fundamental. It is the dependence between consecutive observations that makes it possible to forecast future values and detect when behavior changes.

Time Series as a Discrete Function of Time

Function of Time

The series can be viewed as a mathematical function that maps time instants to real values.

Discrete Time

Time takes on discrete values, such as days, months, or years — not continuous.

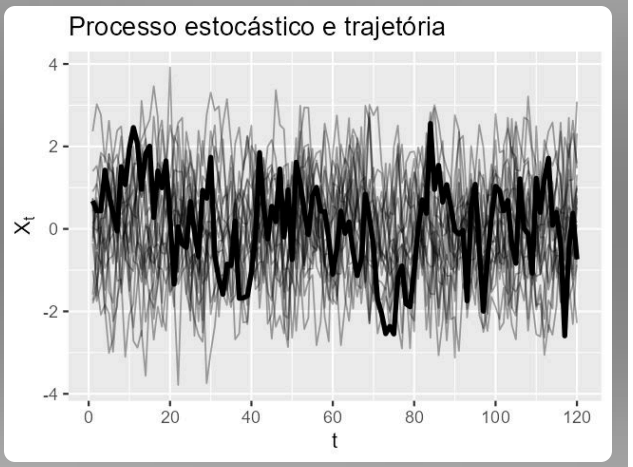
Associated Value

Each instant has a single associated value, forming the temporal sequence.

Mathematically, we define the series as a function:

$$x : \mathcal{T} \rightarrow \mathbb{R}$$

Where $\mathcal{T}$ is the set of time instants (integer values) and $\mathbb{R}$ is the set of real numbers. This functional view allows time series to be treated with mathematical and probabilistic tools.



Stochastic Process and Observed Trajectory

Think of the stochastic process as a generating mechanism we never observe directly — it contains all possible trajectories. The observed series is just one of them: the concrete path the process actually took.

Stochastic Process

$$\{X_t\}_{t=1}^{T}$$

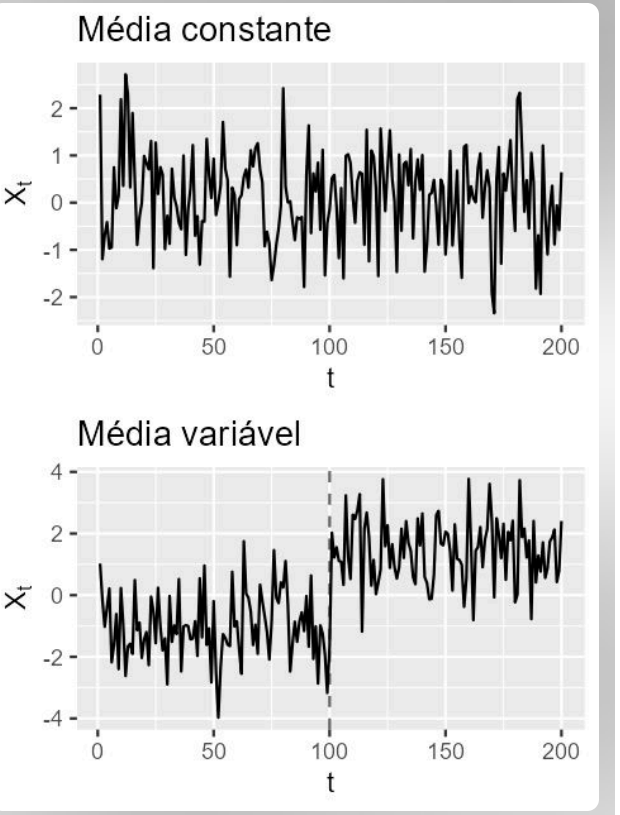
All possible trajectories, described by a probabilistic model.

Observed Trajectory

$$x_t = X_t(\omega)$$

The concrete path the process took — the data we have at hand.

We never have access to the generating process — only to the trajectory it produced. That is why prediction and event detection are inference tasks: from a single observed realization, we attempt to reconstruct the structure of the underlying mechanism and use it to anticipate the future or identify when something relevant has occurred.

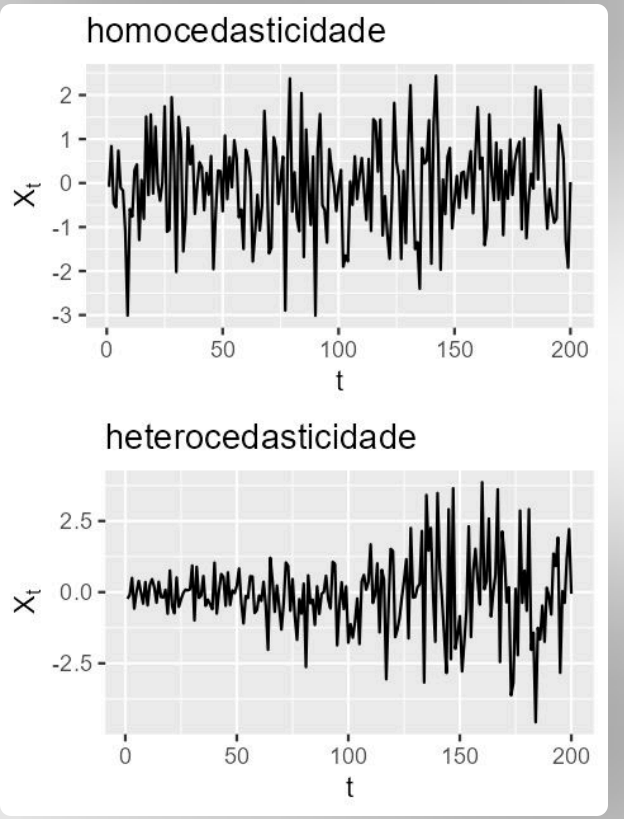


Temporal Mean of the Process

The mean describes the expected level of the series at each instant — it is the reference around which values oscillate. For prediction, the mean defines the baseline the model must anticipate. For detection, persistent deviations from the mean are signals that something has changed.

$$\mu_t = \mathbb{E}[X_t]$$

In practice, we estimate the theoretical mean using the sample mean of the observed series. When this mean does not change over time, the process has a stationary mean — a condition that allows us to use the past as a reliable reference for the future.



Variance of the Temporal Process

If the mean measures the level, the variance measures the uncertainty. It indicates how far values deviate from the expected — and defines the width of prediction intervals. High variance means less precise forecasts; variance that changes over time can itself be a signal of an event.

Theoretical Variance

$$\sigma_t^2 = \mathrm{Var}(X_t)$$

Dispersion of the process at instant t.

Sample Variance

$$s^2 = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})^2$$

Estimate calculated from the data.

When variance changes over time, the series is unstable and harder to model. Constant variance is one of the conditions of stationarity — and its violation may indicate a regime change or a relevant event in the series.

Covariance and Temporal Autocovariance

Mean and variance describe each instant in isolation. Autocovariance goes further: it measures the temporal memory of the process — how much the value at one instant depends on what happened before. This memory is what predictive models exploit to anticipate the future.

Instant t–h

The past value, separated by h periods.

Lag h

The interval between the two instants being compared.

Instant t

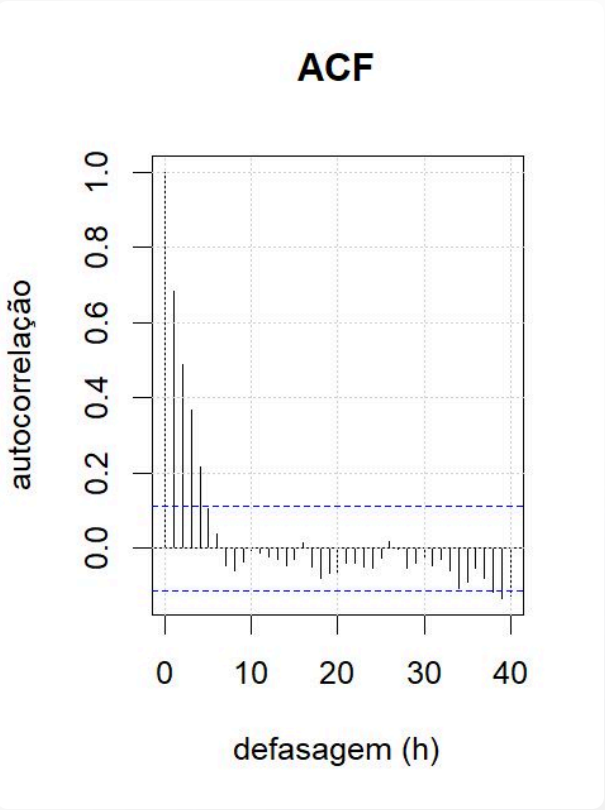
The present value.

The covariance between two instants depends on the interval between them — not on the instants themselves. This allows us to summarize the entire dependence structure in a single function of the lag h, called the autocovariance function:

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t-h})$$

The larger the lag, the smaller the dependence tends to be. Processes with long memory maintain significant autocovariance even for large h.

The function $\gamma(h)$ is the basis for measuring autocorrelation and identifying structural patterns in the series. Changes in this structure over time are potential signals of events — moments when the dynamics of the process have shifted.



Autocorrelation

Autocovariance depends on the scale of the series, which makes comparisons difficult. Autocorrelation solves this: it normalizes dependence to an interval between −1 and 1, making temporal memory comparable across different series and directly interpretable.

Autocovariance

Measures dependence, but in the original data scale.

Normalization

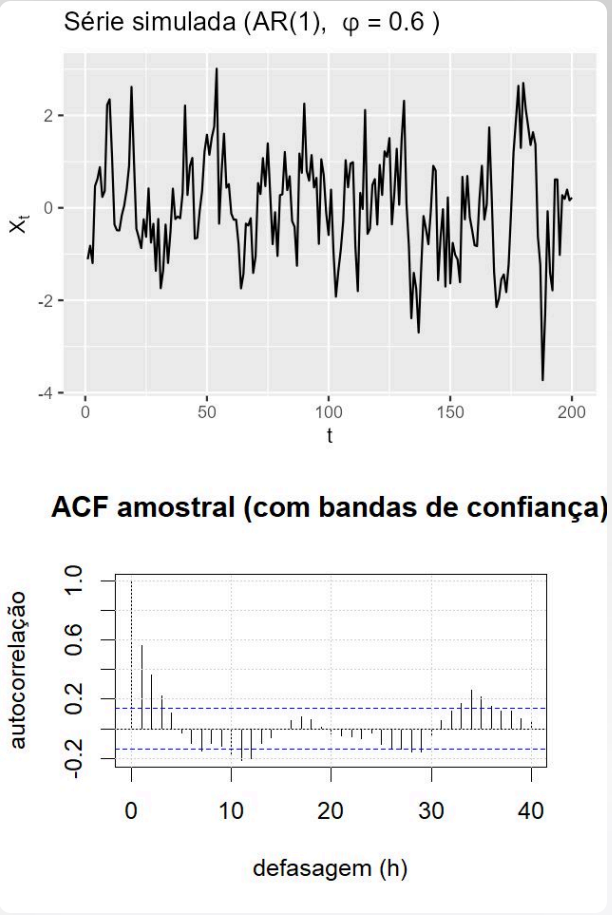
We divide by the variance to eliminate the effect of scale.

Autocorrelation

Standardized result, comparable across different series.

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

Autocorrelation is the primary tool for quantifying the temporal memory of a series. Predictive models such as AR exploit this structure to anticipate future values. Unusual patterns in the ACF — abrupt peaks, breaks, sign changes — can indicate relevant events in the series.



Sample Autocorrelation

The true process is not observable. In practice, we estimate autocorrelation directly from the data — obtaining the sample ACF. It is the starting point for identifying the temporal memory of the series and choosing appropriate predictive models.

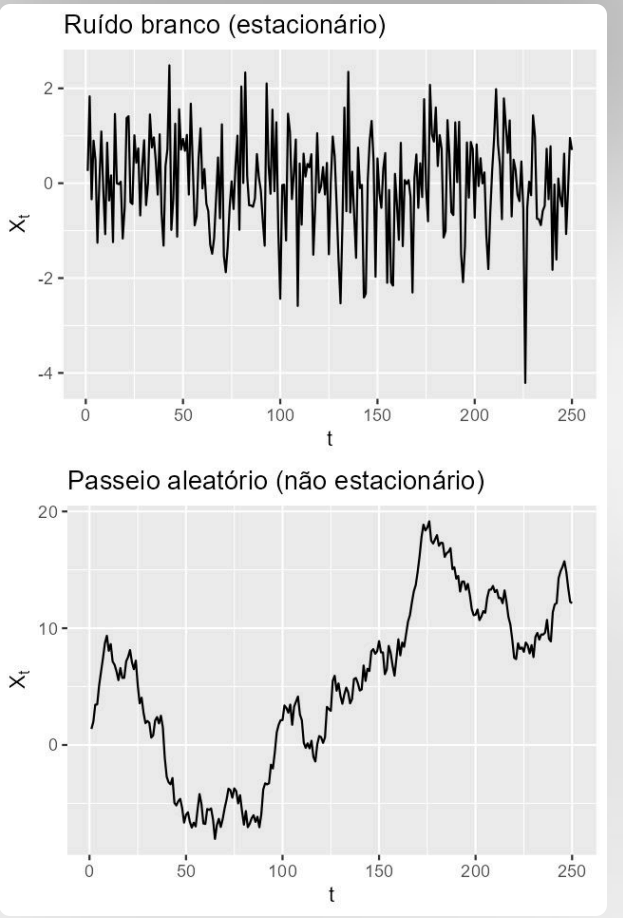
Sample Autocovariance

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=h+1}^{T} (x_t - \bar{x})(x_{t-h} - \bar{x})$$

Sample Autocorrelation

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

The sample ACF reveals the dependence structure present in the data. Prediction models use this structure to anticipate future values. Changes in the ACF over time — detected through rolling windows or statistical tests — are indicators of events that have altered the dynamics of the series.



Stationarity

A series is stationary when its statistical properties — mean, variance, and dependence structure — remain stable over time. Stationarity is not merely a mathematical condition: it is what allows us to use the past as reliable evidence for the future. Without it, what we learn from one period may not apply to another.

Constant Mean

$$\mathbb{E}[X_t] = \mu$$

The mean does not vary over time

Constant Variance

$$\mathrm{Var}(X_t) = \sigma^2$$

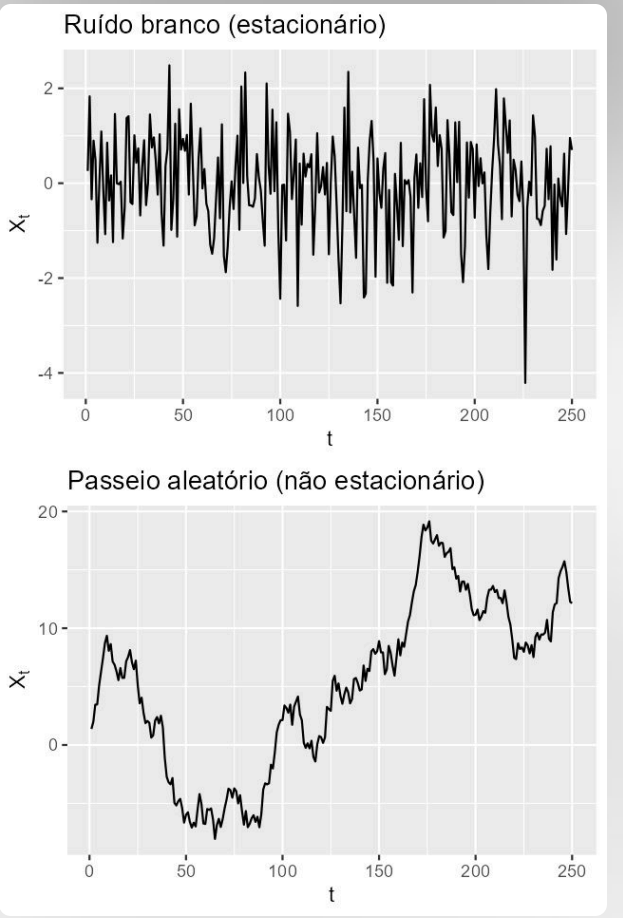
The dispersion remains stable

Autocovariance Dependent Only on Lag

$$\mathrm{Cov}(X_t, X_{t-h}) = \gamma(h)$$

Dependence depends only on the interval, not on absolute time

When a series does not satisfy these conditions, the past is not a reliable guide for the future — and event detection becomes compromised, as there is no stable reference for normal behavior. This is why verifying and ensuring stationarity is the first step before modeling.



Examples: Stationary and Non-Stationary Processes

Stationary Process

White Noise

$$X_t = \varepsilon_t$$

No temporal memory, properties constant over time

Non-Stationary Process

Random Walk

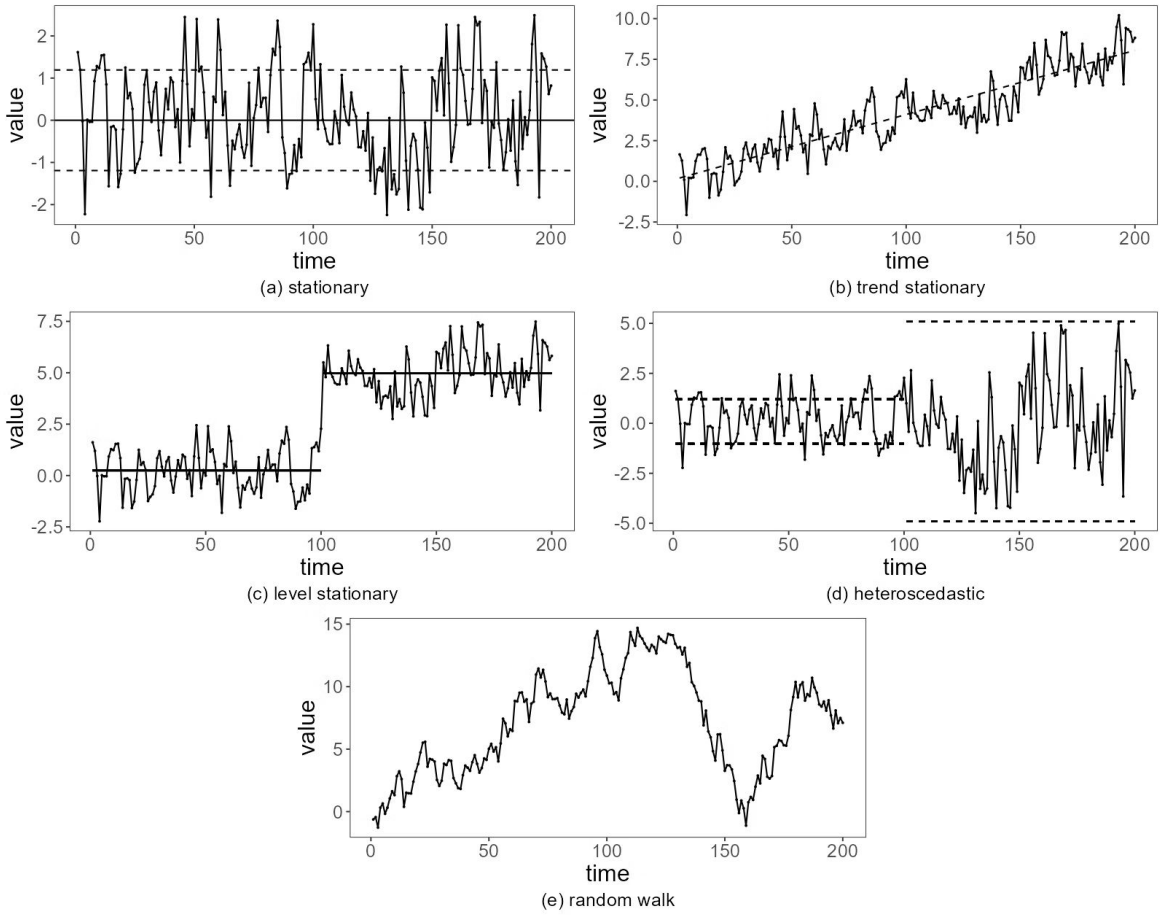
$$X_t = X_{t-1} + \varepsilon_t$$

Accumulates shocks, growing variance over time

Where ε_t is a random term with zero mean and constant variance. White noise represents a simple process with constant properties. The random walk, on the other hand, accumulates shocks, generating growing variance and the absence of stationarity. This contrast helps explain why many real-world series need to be transformed before modeling.

Types of Non-Stationarity

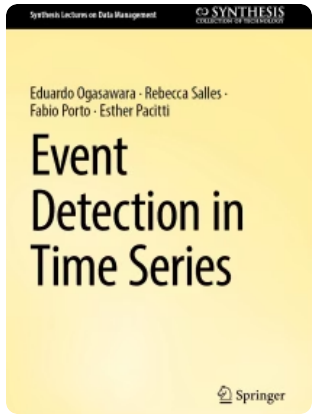
- Deterministic trend shifts the mean over time.
- Structural breaks produce level changes.
- Heteroskedasticity alters the variance.
- Unit root produces difference-stationary behavior.

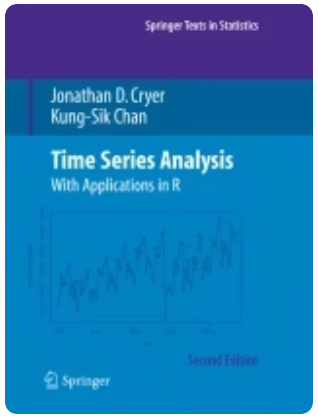


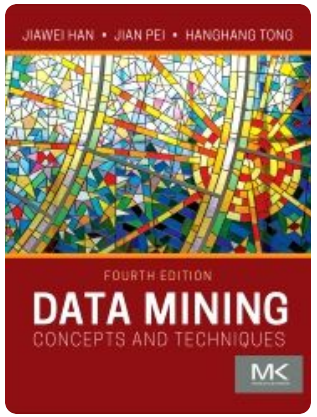
Impact on Event Detection

- Non-stationarity can look like an anomaly.
- Level shifts can be confused with persistent outliers.
- Unaddressed seasonality can generate periodic false positives.
- Proper preprocessing reduces these ambiguities.

Bibliographical References







Event Detection in Time Series

Ogasawara, E.; Salles, R.; Porto, F.; Pacitti, E. Springer Nature Switzerland, 2025.

Time Series Analysis: With Applications in R

Cryer, J. D.; Chan, K.-S. Springer, 2008.

Data Mining: Concepts and Techniques

Han, J.; Pei, J.; Tong, H. Morgan Kaufmann, 4th ed., 2022.